What the professor should be prompted to enter:

- Activity name
- Group size(preferred)
- 5 topics to be entered as a string to be options for preferred topics(later when working on the student side will allow them to choose more than one option here, not a concern for current sprint)
- 5 skills to be entered as a string to be options for (later when working on the student side will allow them to choose more than one option here, not a concern for current sprint)

RULES GOVERNING GROUP FORMATION LOGIC

Hard constraints

A. Student must belong to the course space

A student can only be grouped in a course space they are enrolled in.

B. A student can appear in only one team

Each student can only be assigned once in a single group formation activity.

C. Team size must stay within allowed bounds

Not necessarily exact size, but within:

- min_team_size
- max_team_size

Soft constraints

A. Availability overlap

Students with similar availability should be grouped together when possible. (We will decide exactly what options for students to select later)

B. Topic / project interest match

ALLOW PROFESSORS TO ENTER 5 and ONLY 5 TOPICS TO CHOOSE AS OPTIONS FOR STUDENTS WHEN PROMPTED

C. Skill balance

ALLOW PROFESSORS TO ENTER 5 and ONLY 5 SKILLS TO CHOOSE AS OPTIONS FOR STUDENTS WHEN PROMPTED

D. Leadership balance

Students choose from 5 options, to say how comfortable they are with taking a leadership role(from 1(least confident) to 5(most confident))

E. Balanced team sizes around preferred size

Example:
If preferred size is 4, then:

- 4 is best
- 3 or 5 is acceptable
- the algorithm should prefer solutions closest to 4

F. Diversity or similarity on chosen dimensions

Depending on professor intent, they may want:

- similar students together
  or
- mixed teams

Consider for later iterations

G. Preferred partner requests

If a student says they want to work with someone, the system should try to honor it.

SOURCE MATERIAL USED TO IMPLEMENT GROUP FORMATION LOGIC

To allow space for future development, surveys will sort students based on concrete constraints, and calculations using weighted options (such as a student's preferred skill or topic). These calculations provide an approximation value that can be used to compare *similarities* and *dissimilarities* for each pair of students.

The following source material was consulted to implement the algorithm that sorts students based off parameters:

- **https://help.feedbackfruits.com/hc/en-us/articles/23527084211346-Group-Formation-Algorithm-Explainer**